Checkpoint results and finish plan

Daniel Garcia Mendez | COSC 4371 | Project 11 | Prepared October 29, 2026

This document supplies Sections 3-5 for the checkpoint. Keep the completed dataset citations in your existing document; Section 2(b) is results/audits/checkpoint_2b/audit_table.md and final Sections 2(c)-2(d) evidence is results/splits/section_2c_final.md.

3. Baseline and metrics

(a) Pipeline

Map labels to benign=0 and attack=1; align duration_seconds, forward_packets and backward_packets; convert CICIDS Flow Duration from microseconds to seconds; exclude invalid input rows and apply finalized split assignments. Train only on 68,048 UNSW rows. There is no scaling or imputation. Validation has 17,018 rows; no refit includes validation.

RandomForestClassifier: n_estimators=200, criterion=gini, max_depth=None, min_samples_split=2, min_samples_leaf=2, max_features=sqrt, bootstrap=True, class_weight=None, random_state=42, n_jobs=-1. Other defaults and full settings are recorded in results/baseline/run_metadata.json.

Measured fit time: 2.02 seconds. Scoring/point-metrics time: 68.63 seconds. Hardware: AMD64 Family 23 Model 113 Stepping 0, AuthenticAMD, 12 logical CPUs; Windows-10-10.0.19045-SP0. Training/scoring workflow: 80.26 seconds, excluding the subsequent course-checker run.

(b) Threshold

Positive class: attack (1). Predict attack when score >= 0.9426192528735632. Select the lowest observed validation score threshold whose empirical validation false-positive rate is at most 5%; score ties remain together. Threshold selection uses validation only. Measured validation FPR: 4.9895%; recall: 55.3175%. The same model and threshold score all test datasets.

(c) Checker metrics

Values below are copied from the unchanged course checker; brackets contain 95% row-bootstrap intervals. Default bootstrap seed=0; 1,000 UNSW resamples and 200 per external dataset. The full output is results/baseline/checker_output.txt.

Test set	Test rows	Positive rows (%)	Random AP
UNSW-NB15 official test	82,332	45,332 (55.06%)	0.5506
CICIDS2017	2,830,628	557,646 (19.70%)	0.1970
TON_IoT	211,043	161,043 (76.31%)	0.7631

Test set	Precision [95% CI]	Recall [95% CI]	FPR [95% CI]
UNSW-NB15 official test	0.7005 [0.6927, 0.7085]	0.2084 [0.2048, 0.2120]	0.1092 [0.1059, 0.1126]
CICIDS2017	0.3220 [0.3207, 0.3233]	0.3007 [0.2994, 0.3022]	0.1553 [0.1549, 0.1558]
TON_IoT	0.8627 [0.8606, 0.8645]	0.5335 [0.5311, 0.5357]	0.2735 [0.2704, 0.2772]

Test set	AP [95% CI]	AUROC [95% CI]
UNSW-NB15 official test	0.7530 [0.7489, 0.7576]	0.7833 [0.7799, 0.7867]
CICIDS2017	0.2829 [0.2818, 0.2841]	0.5048 [0.5039, 0.5060]
TON_IoT	0.7943 [0.7921, 0.7962]	0.5169 [0.5139, 0.5191]

(d) Random reference and interpretation

UNSW-NB15 official test: AP exceeds its random-scorer reference and AUROC exceeds 0.5.
CICIDS2017: AP exceeds its random-scorer reference and AUROC exceeds 0.5.
TON_IoT: AP exceeds its random-scorer reference and AUROC exceeds 0.5.

AUROC drops from UNSW by 0.2784 on CICIDS2017 and 0.2663 on TON_IoT. External ranking is close to random despite above-random point estimates. This three-feature baseline does not demonstrate reliable cross-environment transfer. Collector differences and removal of 51.49% of original UNSW training-source rows

limit attribution to environment alone.

Precision and AP depend on prevalence; AUROC is the primary comparison. A validation FPR cap does not constrain test FPR. Repeated flows may violate row-bootstrap independence, so the checker's intervals do not quantify every source of uncertainty.

(e) Too-good check and warnings

None of precision, recall, AP or AUROC reaches 0.99 on any test set. Final split checks report zero exact training/test and training/validation matches and zero protected near-group crossings. Host/session independence cannot be checked with the prepared UNSW identifiers.

Checker warnings:

None.

(f) Reproducibility

Repository: https://github.com/DanielGarciaMendez/Security-Analytics-Network-Intrusion-Detection

Python 3.13.7; exact installed packages are recorded in results/baseline/environment.txt. Seeds: split/model=42, course bootstrap=0. Data download locations and file paths are documented in README.md; raw datasets are not committed.

One command from repository root after installing dependencies and placing raw downloads:

```powershell
.\.venv\Scripts\python.exe src\run_checkpoint.py
```

The complete command was executed successfully in an isolated source/output workspace using the existing locked environment and original raw inputs. Assignments, fitted tree arrays and all thresholded test predictions reproduced. Probabilities differed by at most 3.3306690738754696e-16; tiny score-tie changes slightly altered rank metrics, so the original saved scores remain the source of this document's numbers. See results/reproduction/verification.json. A fresh package installation, fresh Git clone and independent data downloads were not tested. Latest artifacts still need to be committed and pushed.

4. Figure

Figure: Random Forest on the UNSW-NB15 official test set; 82,332 rows and 55.06% attacks. Cells show counts; rows are true labels and columns are predicted labels.

At the validation-selected threshold, the baseline detects 20.84% of UNSW test attacks at a 10.92% false-positive rate, showing that even source-dataset operating performance differs from validation.

5. Plan to finish

(a) Required elements

Proposed dates based on the work remaining on October 29, 2026; these are planning targets, not an assumed course deadline. Scope follows the available proposal (Random Forest and XGBoost); reconcile any additional instructor-assigned requirements separately.

Required element	Status	Completion/target date
Three datasets, binary labels and shared feature mapping	done	October 28, 2026
Audit, exclusions, group-aware splits and leakage checks	done	October 28, 2026
Recent dataset-paper citations	done (user-confirmed)	October 29, 2026 or earlier
Random Forest training, validation threshold and three test sets	done	October 28, 2026
Course checker, intervals, figure and results interpretation	done	October 29, 2026
XGBoost baseline on the same features and assignments	not started	October 30, 2026
XGBoost validation-only threshold, test scores and checker	not started	October 30, 2026
Cross-dataset comparison for both classifiers	started	October 31, 2026
Limitations, final report and AI-use disclosure	started	November 1, 2026

| Fresh-copy reproduction, repository publication and final review | started (isolated rebuild verified) | November 2, 2026 |

(b) Main comparison

Variable: test dataset (UNSW-NB15 official test, CICIDS2017, TON_IoT). For each classifier, hold training rows, preprocessing, representation, fitted model and validation-selected threshold fixed. Use the same UNSW validation assignments for RF and XGBoost; each model selects its own threshold with the same <=5% validation-FPR rule. Report each model as a separate results-table row. No test-driven retuning is planned.

Primary metric: AUROC. Operational question: does source-to-external AUROC fall by a meaningful amount? Let drop = UNSW AUROC minus external AUROC. Count yes when drop >= 0.050, no meaningful decline when drop <= 0, and a smaller positive drop as inconclusive under this decision margin. Apply the rule separately to each external dataset and each classifier. The widest printed baseline AUROC interval has half-width 0.0034; the 0.050 yes margin exceeds it. This rule is specified after observing the baseline, so it is a planning rule for remaining work, not a preregistered confirmatory hypothesis.

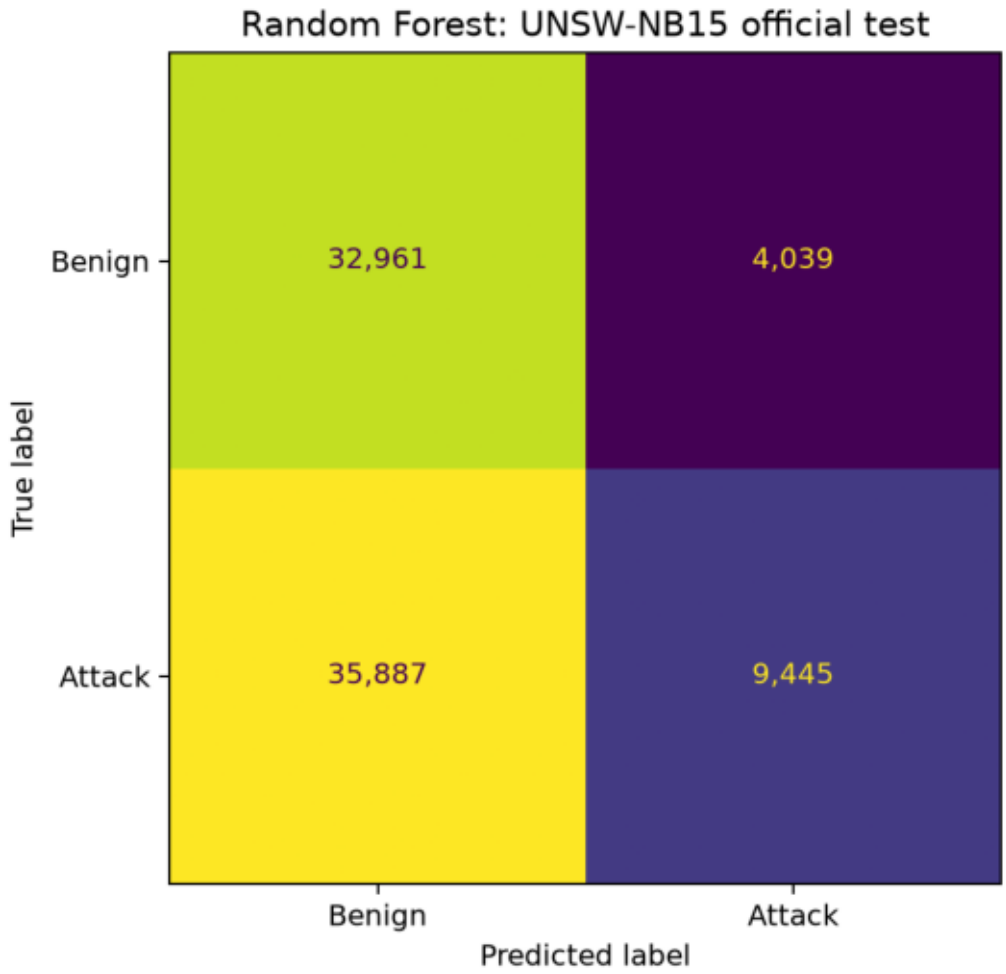
Classifier	UNSW test AUROC	CICIDS2017 AUROC	TON_IoT AUROC
Random Forest	0.7833 [0.7799, 0.7867]	0.5048 [0.5039, 0.5060]	0.5169 [0.5139, 0.5191]
XGBoost	pending	pending	pending

(c) Compute budget

Measured RF fit: 0.0337 min; scoring and point metrics: 1.14 min; course checker: 5.73 min. Budget one XGBoost fit at 10x RF fit (0.34 min, a provisional multiplier, not a measurement), plus one XGBoost scoring/checker pass and one RF reproduction pass. Estimated remaining compute = 11 x RF fit + 2 x RF scoring time + 2 x checker time = 14.12 min. Reserve twice that (28.23 min) for variation; this excludes downloads, repeated full audits and writing. Replace the XGBoost estimate with its measured runtime after the first run.

AI-use disclosure draft

OpenAI Codex assisted with data-audit, feature-alignment, splitting and Random Forest pipeline code; tests; interpretation; and preparation of this results/plan draft. Course-checker output supplies the reported metrics. The author should review the work and format this disclosure according to the syllabus; the syllabus's exact citation requirements were not supplied.



UNSW-NB15 official test: 82,332 rows; 55.06% attacks.
At the validation-selected threshold: recall 20.84%, FPR 10.92%.